



Multi-Criteria Optimization of Electric Vehicle Charging and Route Planning

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1. Motivation & Problem Definition

Commercial navigation tools evaluate shortest paths statically, ignoring EV battery physics, charging curves, and weather. Academic models focus on corporate *fleets*, ignoring individual drivers.

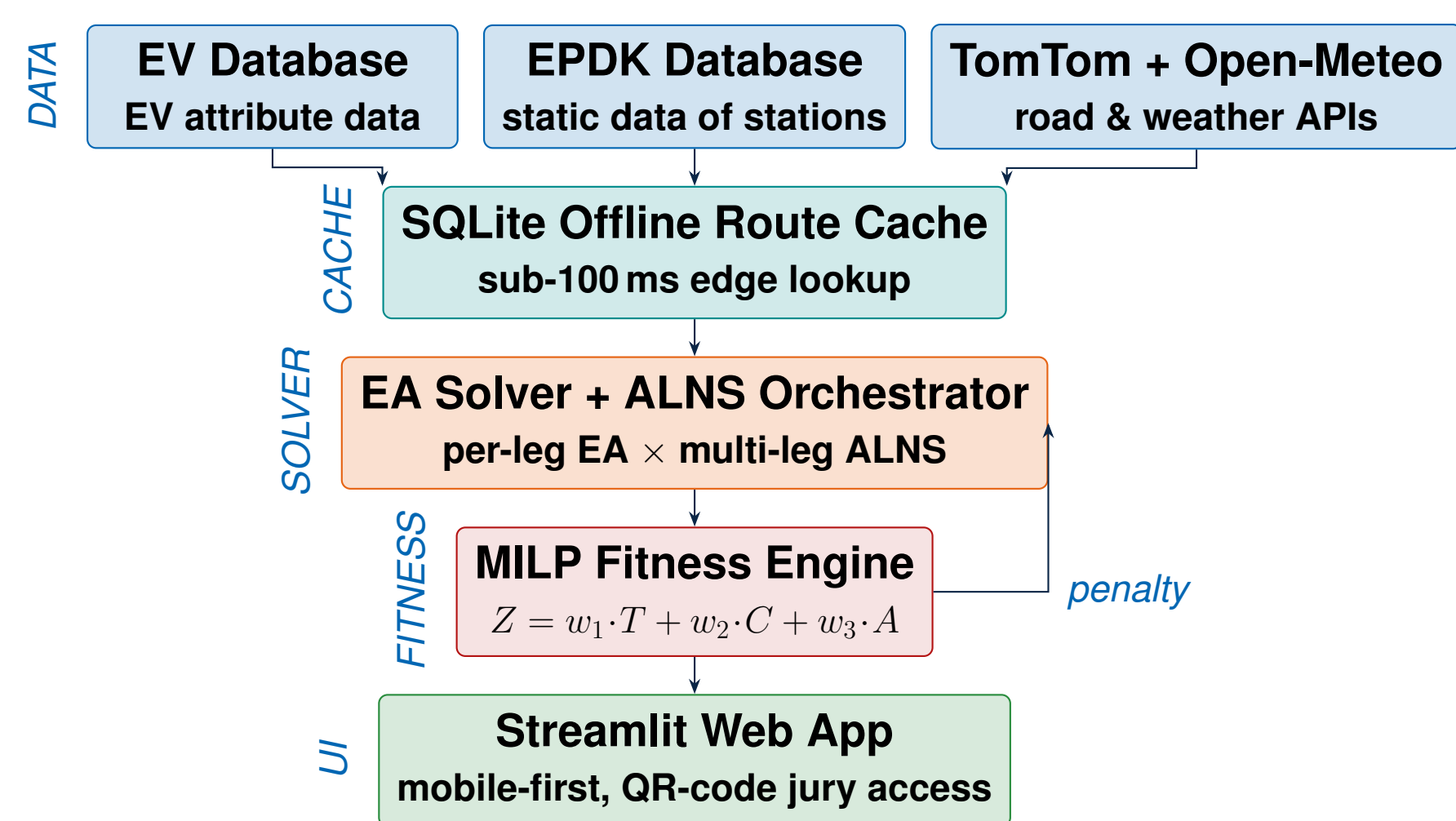
We formulate the **Multi-Criteria EVCRP** to personalise routes based on three user-tuneable priority weights:

- **Time** (w_1): Driving + queue + non-linear charging delay.
- **Cost** (w_2): Financial tariffs across charging operators.
- **Anxiety** (w_3): Safety buffers above critical battery floors.

Our global objective minimizes the weighted criteria:

$$Z = w_1 \cdot \text{Time} + w_2 \cdot \text{Cost} + w_3 \cdot \text{Anxiety}$$

2. System Architecture



3. Data Modalities

- **Live Streams:** TomTom routing & Open-Meteo weather (F_{ext}).
- **Static Caches:** EPDK database (14,825 plugs) + EV spec tables.
- **Offline SQLite:** Pre-fetched road edges, enables sub-3 s solver.
- **Mock Metrics:** Dynamic pricing (c_q) & queue delays $L_i(t)$.
- **Stress Layers:** Seeded randomised weather fields.

4. EA + ALNS Hybrid Solver

Tier 1 — Per-leg Evolutionary Algorithm (EA)
Chromosome encodes routing sequence, charge amounts and a continuous **velocity gene** v_i :

$$[w_0 \rightarrow (s_1, q_1, \mathbf{v}_1) \rightarrow \dots \rightarrow (s_n, q_n, \mathbf{v}_n) \rightarrow w_K]$$

$v_i \in [0.7, 1.3]$; consumption scales as $E = E_{eco} \cdot v_i^{1.6}$ (aero-drag model); speed capped by TomTom real-world limits.

Tier 2 — Multi-leg ALNS Orchestrator (pipeline.py)
Manages SOC hand-offs between legs, ensuring global feasibility across all waypoints via Adaptive Large Neighbourhood Search.

EA Parameter	Value
Population P	60 individuals
Init strategy	20% greedy + 80% random
Max generations	300 (early-stop: 100)
Selection	Tournament $k=3$, Elitism $E=2$
Crossover rate	0.85 (multi-point leg exchange)
Structural mut.	0.15 (insert/delete station)
Parameter mut.	Gaussian $\Delta q_i, \Delta v_i$

5. UN Sustainable Development Goals

Impact Area	Type	Alignment & Project Definition
Social (SDG 10)	Direct/Ind.	Democratizes routing optimization for individual drivers, reducing technological gaps.
Health (SDG 3)	Direct/Ind.	Reduces cognitive workload and driver stress by mitigating range anxiety.
Security (SDG 16)	Direct	Securely processes user location and battery telemetry via trusted APIs.
Economic (SDG 8)	Direct/Ind.	Optimizes financial trip costs and supports EV charging sector growth.
Sust. & Env. (SDG 11–13)	Direct	Maximizes energy efficiency via partial charging, weather-aware routing, and CO ₂ reduction.

6. Mathematical Model Formulation

Sets & Parameters	Decision Variables
$W = \{w_0, \dots, w_K\}$: Mandatory way-points	$x_{ij} \in \{0, 1\}$: Edge traversal flag
S : Candidate charging stations	$v_i \in \{0, 1\}$: Station charging event active
$N = W \cup S$: Node set, $E \subseteq N \times N$: Edge set	$q_i \geq 0$: Energy percentage charged at node i
$B_{\max}$: Battery capacity (100%), B_{ceil} : Charging ceiling SOC	$y_i \in [0, B_{\max}]$: Battery SOC upon arrival at i
B_{floor} : Safety floor SOC, B_{start} : Initial SOC	$T_{\text{arr},i}, T_{\text{dep},i}$: Arrival / departure times
τ : Anxiety threshold, M : Big-M scalar	$w_i \geq 0$: Time-dependent queue latency at i
e_{ij} : Edge energy consumption	$p_i \geq 0$: SOC anxiety penalty variable
t_{ij} : Edge travel time, c_q : Unit charging cost	
r_i : Unit charging duration, $L_i(t)$: Queue delay	
w_1, w_2, w_3 : Weights (time, cost, anxiety)	

Multi-Objective Function:

$$\min Z = w_1 \left(\sum_{(i,j) \in E} x_{ij} t_{ij} + \sum_{i \in N} (q_i r_i + w_i) \right) + w_2 \left(\sum_{i \in N} q_i c_q \right) + w_3 \left(\sum_{i \in N} p_i \right)$$

Constraints & Mathematical Dissection:

• **Multi-Leg Network Flow Conservation:**

$$\begin{aligned} \sum_{j: (w_0, j) \in E} x_{w_0, j} &= 1, & \sum_{i: (i, w_K) \in E} x_{i, w_K} &= 1 \\ \sum_{j: (w_k, j) \in E} x_{w_k, j} &= \sum_{i: (i, w_k) \in E} x_{i, w_k} = 1 & \forall w_k \in W \setminus \{w_0, w_K\} \\ \sum_{j: (i, j) \in E} x_{ij} &= \sum_{j: (j, i) \in E} x_{ji} \geq v_i & \forall i \in S \end{aligned}$$

Ensures path starts at w_0 , ends at w_K , visits all waypoints, and restricts station activation v_i .

• **Two-Sided SOC Tracking (Energy Sandwich):**

$$y_j \leq y_i + q_i - e_{ij}(v) + M(1 - x_{ij}) \quad \forall (i, j) \in E$$

$$y_j \geq y_i + q_i - e_{ij}(v) - M(1 - x_{ij}) \quad \forall (i, j) \in E$$

Collapses to exact physics $y_j = y_i + q_i - e_{ij}(v)$ when $x_{ij} = 1$, preventing artificial SOC values.

• **Velocity Dynamics & Aerodynamic Drag:**

$$t_{ij}(v) = \frac{d_{ij}}{v} \cdot 60 + \Delta t_{\text{traffic}}, \quad e_{ij}(v) = \left(\frac{d_{ij} \cdot R(v)}{C_{\text{bat}}} \right) \cdot F_{\text{ext}}$$

$$R(v) = R_{\text{eco}} \cdot \left(\frac{v}{v_{\text{eco}}} \right)^{1.6}, \quad v \in \{70, 90\} \text{ km/h}$$

Calculates speed-dependent travel durations and non-linear aerodynamic energy consumption.

• **Monotonic Clocks on Cyclical Networks:**

$$T_{\text{arr},j} \geq T_{\text{arr},i} + t_{ij}(v) + q_i r_i + w_i - M(1 - x_{ij}) \quad \forall (i, j) \in E$$

$$y_i^{n+1} = y_i^n - (e_{ik} + e_{ki}) + q_k < y_i^n \quad (\text{if cycle occurs})$$

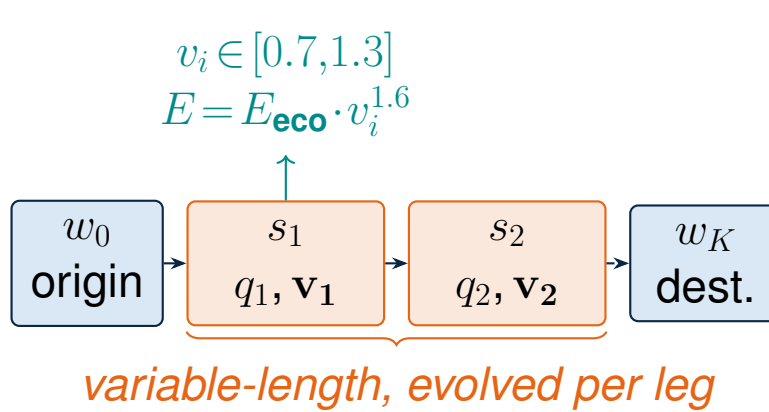
Prevents infinite cycles on real maps without forcing a Directed Acyclic Graph (DAG) topology.

• **Queue Latency & Driving Anxiety Penalty:**

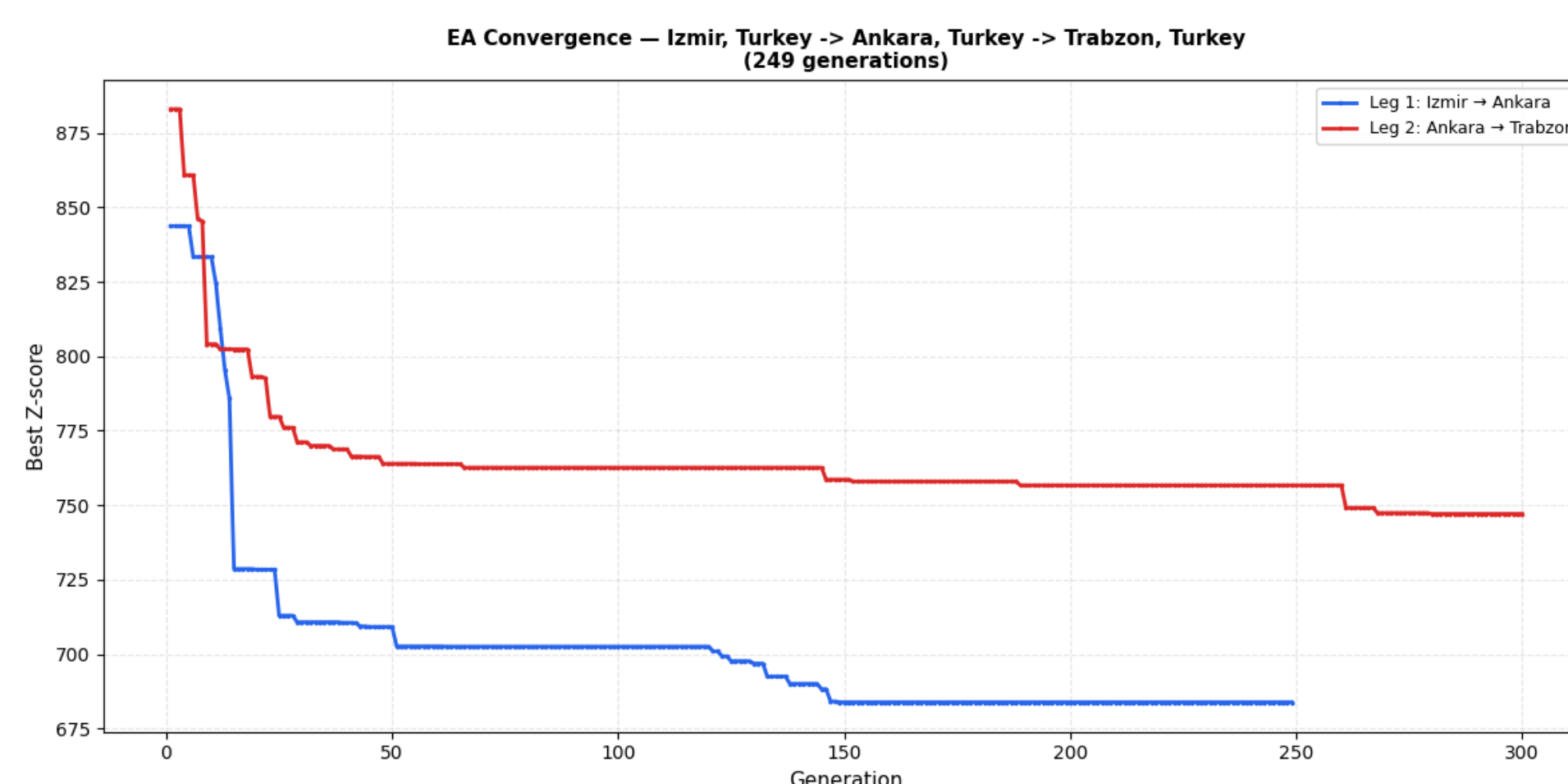
$$w_i \geq L_i(T_{\text{arr},i}) \cdot v_i \quad \forall i \in S, \quad p_i \geq \tau - y_i, \quad p_i \geq 0 \quad \forall i \in N$$

Enforces time-dependent queue delay functions and linearized state-of-charge anxiety penalties.

Chromosome Encoding:



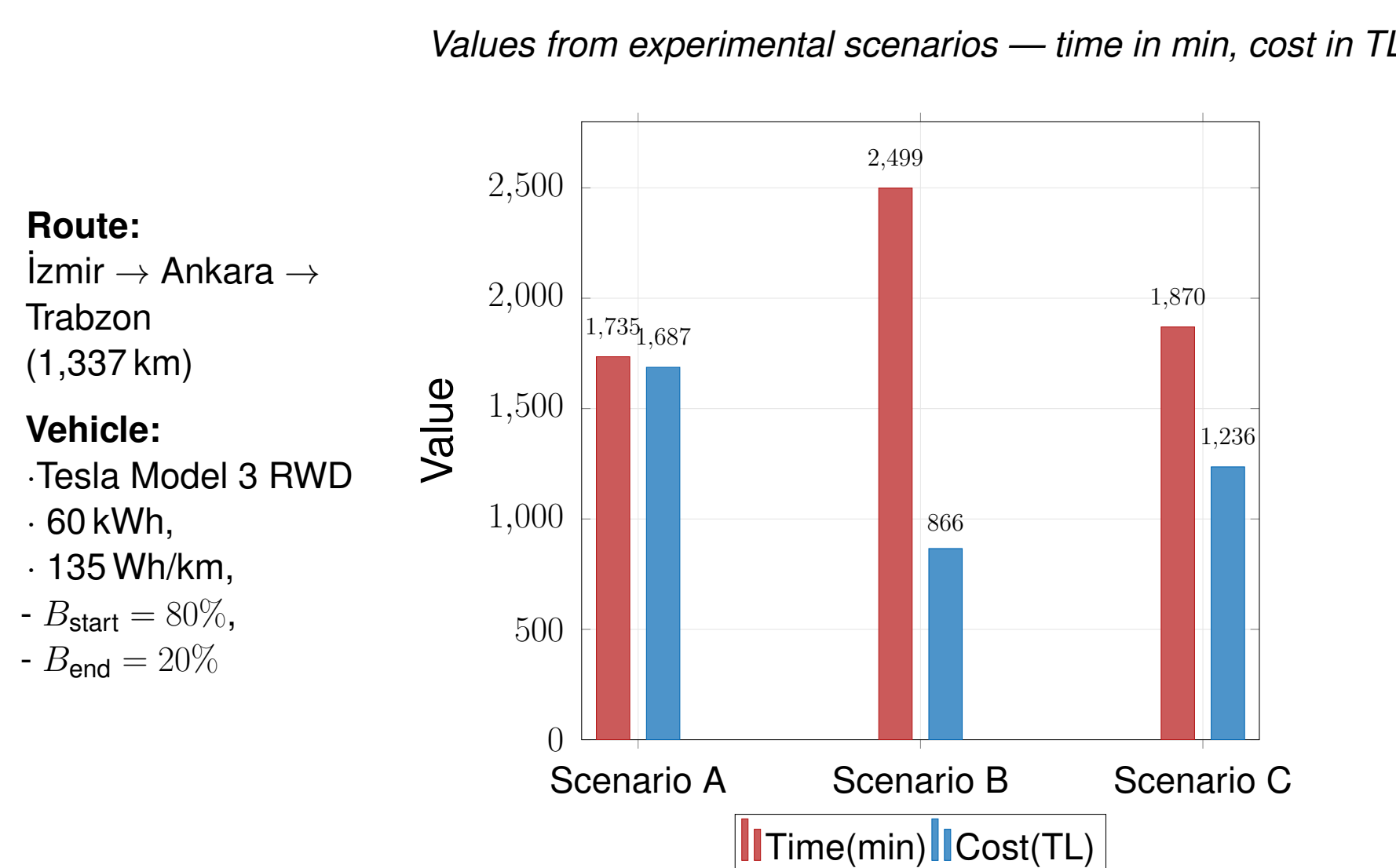
EA Convergence — Z-Score vs. Generation:



No DAG Restriction. Absolute chronological arrival times $T_{\text{arr},i}$ replace the traditional $i < j$ forward-ordering constraint. Since $e_{ij} > 0$ for every real road segment, any cycle degrades SOC monotonically — infinite loops are *physically* eliminated without restricting graph topology.

7. Experimental Results

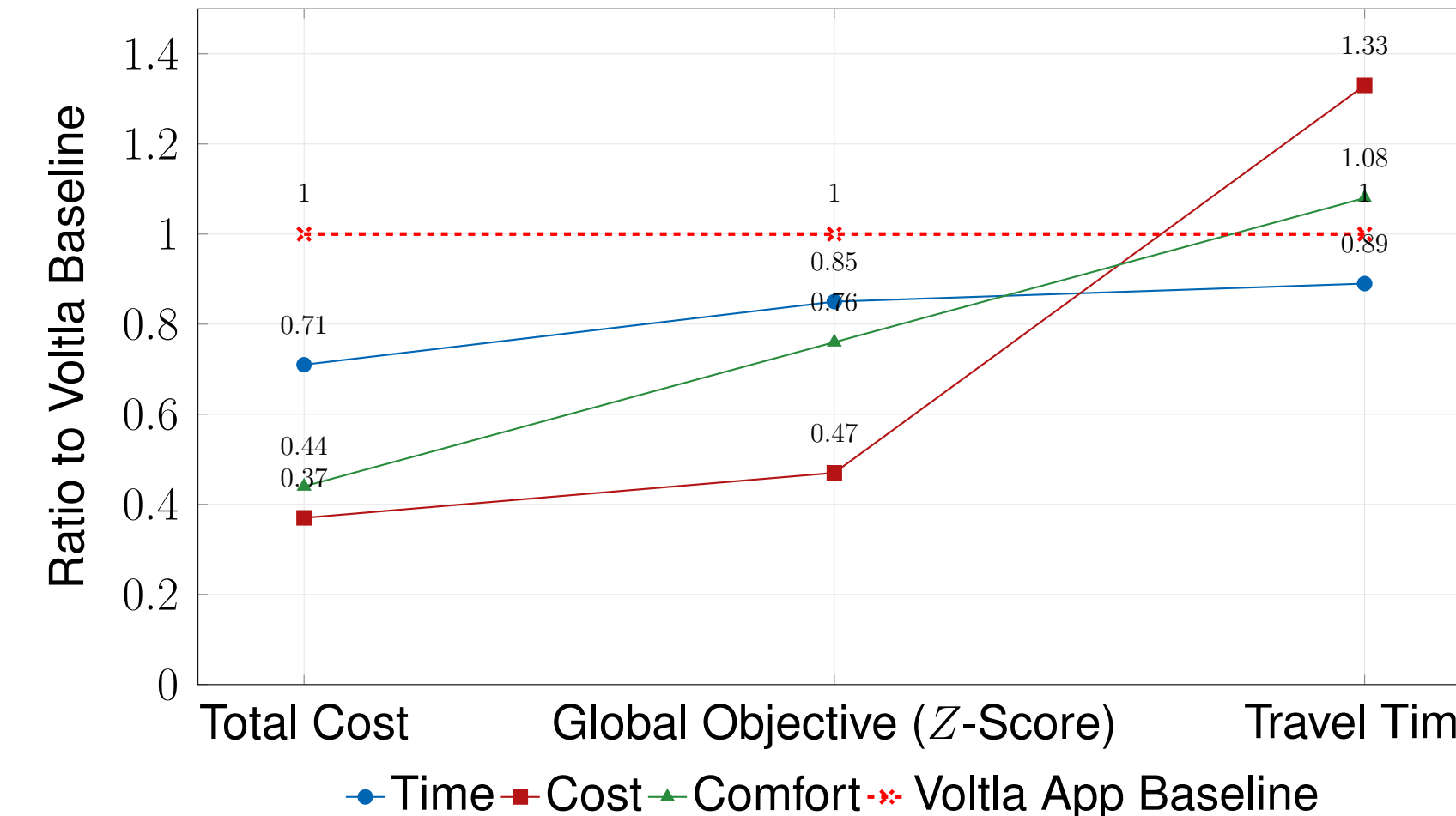
Scenario Comparison — Time vs. Cost:



Note: Scenario B optimizes strictly for travel cost, Scenario A prioritizes travel speed, Scenario C prioritizes comfort and energy efficiency.

Performance Comparison against Voltla App

Benchmark against Voltla App routing engine across the same multi-leg itinerary.



Note: Our model includes external variables to evaluate realistic performance.

8. User Interface

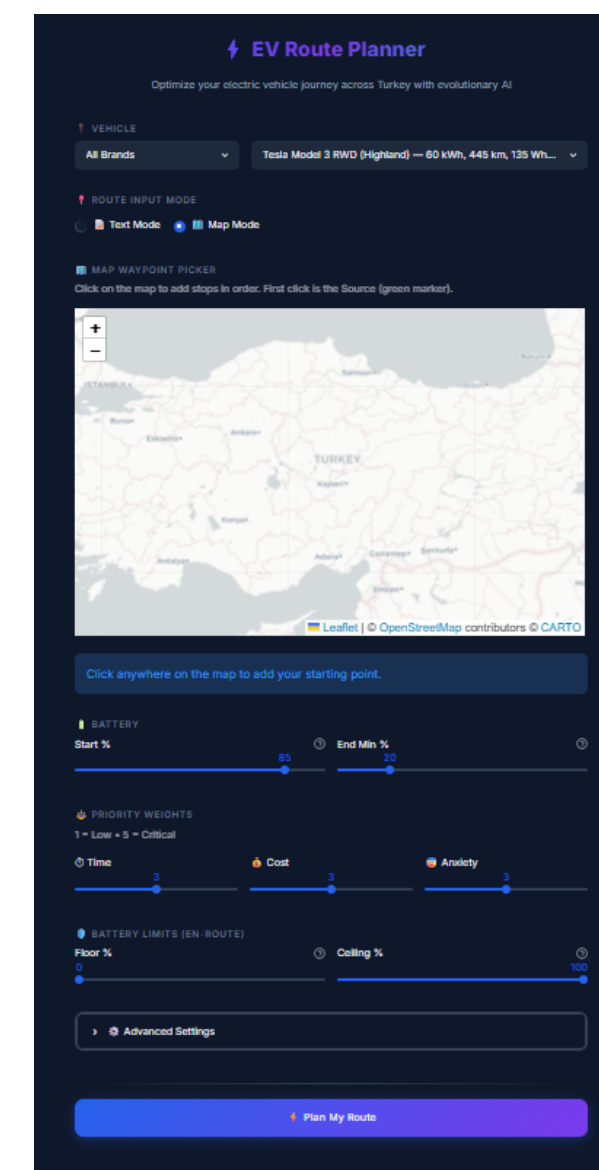


Figure 1: Input Area

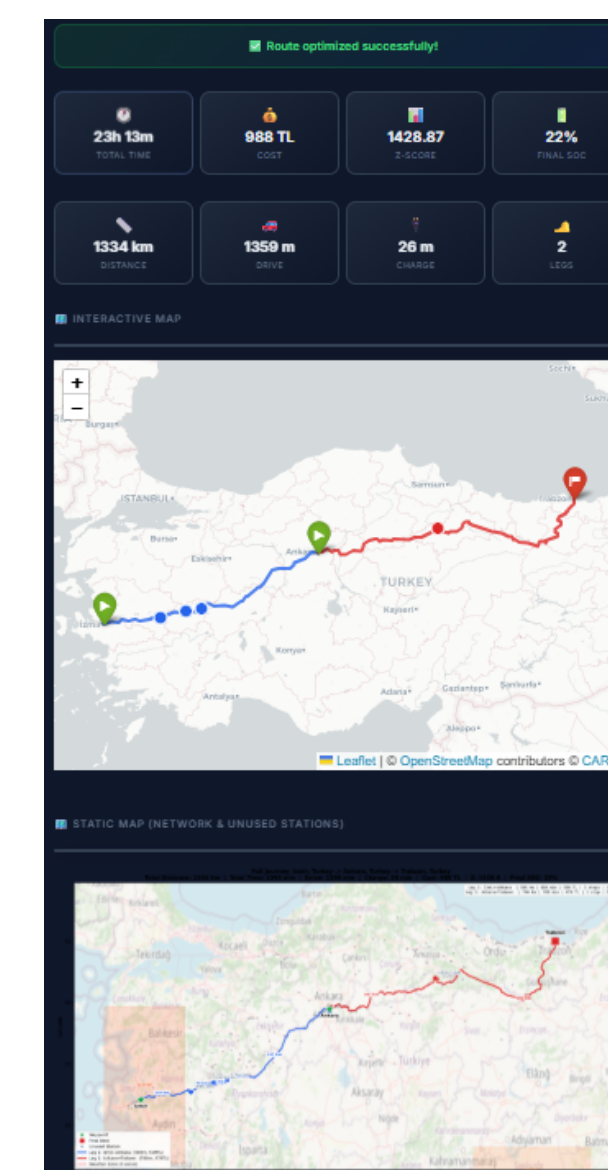


Figure 2: Result Area

9. Conclusions & Future Work

- **True network topology:** Chronological time-tracking replaces DAG constraints, enables real-world loops and back-tracking.
- **EA + ALNS hybrid:** Sub-3 s multi-leg optimisation, viable for live consumer web requests.
- **Individual-centric:** First framework targeting individual drivers (not fleets) with full UI personalisation.
- **Performance gains:** Outperforms commercial baseline in all scenarios, achieving up to **63% financial cost savings** and **53% global objective (Z-score) improvement** (under Scenario B).
- **SDG alignment:** Advances **SDG 11** (Sustainable Cities) and **SDG 13** (Climate Action).

Future work:

- Live IoT station-occupancy telemetry replacing mock $L_i(t)$.
- Real-time Open-Meteo weather replacing randomised stress fields.
- Non-linear battery thermal throttling & degradation models.
- Game-theoretic multi-vehicle fleet coordination.

10. Acknowledgements

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